

JEE MAIN REVISION MASTER: S-BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	S Block Elements — Core Principles & Governing Laws	7 PYQs	Core Principle
Standard	S Block Elements — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics
Standard	S Block Elements — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
S Block Elements — Core Principles & Governing Laws	$\text{Al}_2\text{O}_3, \text{K}_2\text{O}, \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
S Block Elements — Applied Cases, Reactions & Traps	$\text{LiNO}_3, \text{NO}_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 S Block Elements — Core Principles & Governing Laws (Moderate Yield — 7 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in S Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in S Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in S Block Elements PYQs.

Representative Source PYQs:

- [2021 Shift] Q#2 (MCQ): "The property of the alkaline earth metals that increases with their atomic number is..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "The oxide of potassium that does not exist is..." [Ans: N/A]
- [2020 Shift] Q#4 (MCQ): "The metal that produce H_2 with both dil. HCl and NaOH(aq) is..." [Ans: N/A]

3.2 S Block Elements — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of S Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following nitrate will decompose to give NO_2 on heating ?..." [Ans: N/A]

3.3 S Block Elements — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in S Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following nitrate will decompose to give NO_2 on heating ?..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "The property of the alkaline earth metals that increases with their atomic number is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which one of the following nitrate will decompose to give NO_2 on heating ?

- (A) NaNO_3
- (B) KNO_3
- (C) RbNO_3
- (D) LiNO_3

Correct Answer: None

Solution Analysis: Among the given nitrates, only LiNO_3 will decompose to give NO_2 when heated. In contrast, the nitrates of sodium (Na), potassium (K), and rubidium (Rb) decompose to release oxygen instead. The decomposition reactions are as follows: For lithium nitrate (LiNO_3): $2\text{LiNO}_3 \rightarrow 2\text{Li}_2\text{O} + 4\text{NO}_2 + \text{O}_2$ For sodium, potassium, and rubidium nitrates (MNO_3), where $\text{M}=\text{Na}, \text{K}, \text{Rb}$): $2\text{MNO}_3 \rightarrow 2\text{MNO}_2 + \text{O}_2$ Here, LiNO_3 uniquely produces NO_2 along with the metal oxide and oxygen.

Benchmark Problem 2: KCET 2021 (Q#2)

The property of the alkaline earth metals that increases with their atomic number is

- (A) ionisation enthalpy
- (B) electronegativity
- (C) solubility of their hydroxide in water
- (D) solubility of their sulphate in water

Correct Answer: None

Solution Analysis: The solubility of the ionic compounds (in this case is hydroxide of alkaline earth metals) depends on two factors, which are (i) lattice energy and (ii) hydration energy. Hence, solubility of alkaline earth metal increases with increase in their atomic number. Whereas, other given properties i.e. ionisation energy, electronegativity and solubility of their sulphate in water decreases with atomic number.